Manikanda Prabhu Bagavathi Krishnan

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Summary

Data analyst and machine learning engineer finishing a Master of Data Science at RMIT University in December 2026, with close to two years of hands-on SQL and Python work. At Smartfusion I analysed 1M+ operational records, built Python pipelines that lifted reporting accuracy by 20% and set up the data models behind Power BI and Tableau dashboards for five business units. At Focus Bear I built a wearable breathing and posture tracker with the CEO and a multi-model AI classifier that reached 94.7% accuracy, and on my RMIT capstone I led data processing for an AI pipeline backed by 122 automated tests. I care most about data people can trust, and I enjoy explaining it to people who don't work with it every day.

Right to work: Student Visa (subclass 500), moving to a Temporary Graduate Visa (subclass 485) with full work rights. No sponsorship required.

Experience

Focus Bear **Apr 2026 – Present**
Machine Learning Engineer Intern *Melbourne, VIC*

- Breathing & Posture Detection (Wearable MVP), Jul 2026 to Present
- Delivered a working end-to-end dual-IMU breathing and posture detection MVP in direct collaboration with the CEO, from M5Stack StickS3 hardware through a Python signal-processing engine to a macOS menu-bar app
- Diagnosed and fixed a posture-calibration bug where the "upright" baseline auto-captured from an arbitrary sensor position, causing systematically incorrect readings, and replaced it with a user-triggered calibration flow verified against real hardware
- Resolved a firmware flashing failure on ESP32-S3 hardware caused by an incompatible reset handshake, implementing an alternative esptool-based flashing procedure to unblock hardware deployment
- Built and iterated on a macOS menu-bar app (Python, rumps, AppKit), redesigning the calibration screen after CEO feedback into a single instruction and button so posture is scored from a pose the wearer chooses, verified on-device
- Validated dual-IMU (BMI270) breathing signal accuracy through structured hardware testing (USB serial + Bluetooth LE streaming), confirming correct chest/belly breathing differentiation
- Attention Score Prediction (Vision-Language Model Classifier), Apr 2026 to Jun 2026
- Achieved 94.7% consensus accuracy on a 58-frame labelled test set by engineering a multi-model ensemble classifier using GPT-5.4 and Gemini 3.1 Flash Lite with Claude Sonnet as tiebreaker, outperforming each individual model at 91.4%
- Improved category accuracy from 72% to 91.4% across 6 autonomous prompt-tuning iterations using Claude Code, eliminating manual trial-and-error
- Engineered separate prompt architectures for GPT and Gemini vision models to classify user attention states from screen and workspace images, enabling both models to feed one consensus classifier
- Resolved a critical Gemini label normalisation bug, building a normalisation function that flagged and corrected invalid responses before scoring
- Built a consensus classifier pipeline achieving 98.3% agreement rate across both models, automatically flagging conflict frames for manual review
- Delivered a clean GitHub PR with full documentation, enabling a smooth handover

Spectrum Tuition Digital (RMIT University Capstone) **Aug 2026 – Nov 2026**
Machine Learning Engineer, Data Processing Lead *Melbourne, VIC*

- Capstone project with a five-person team building an AI pipeline that checks K-12 maths coursebooks before they go to print
- Built an answer checker for maths coursebooks that confirmed 192 of 1,000 Year 8 questions without human review, up from 123, with no mismatches reported
- Designed a validator that sorts each question into one of 8 types, including arithmetic, fractions, multiple choice, time and Roman numerals. It uses exact fraction arithmetic in Python and JavaScript, and sends anything uncertain to a human reviewer
- Wrote 122 automated tests (51 JavaScript and 71 Python) and compared results before and after every change on real coursebook data, so no earlier confirmed answer was lost
- Found and fixed bugs that only showed up on real content, such as typographic symbols being silently ignored and a multiple choice error that could have marked a correct answer wrong
- Reviewed teammates' code by running their branches, documented gaps in the extracted data for the team, and built a browser tool so non-developers can run the checks

Smartfusion Corporate Solutions Pvt Ltd **Feb 2024 – Feb 2025**
Data Analyst *Chennai, India*

- Analysed 1M+ operational records using Python and SQL, applying statistical analysis including significance tests to identify

trends and bottlenecks across five business units, supporting a process redesign that reduced manual processing overhead by 30%

- Built automated ETL data validation and cleaning pipelines in Python, improving reporting accuracy by 20% and strengthening data quality across operational dashboards
- Handled the data modeling behind interactive Power BI and Tableau dashboards, setting up the tables and relationships, so five business units could monitor real time KPIs with fewer decision making delays

Access Healthcare **Nov 2021 – Apr 2023**
Client Partner *Chennai, India*

- Analysed large scale healthcare claims datasets to identify approval bottlenecks, and implemented structured tracking workflows that increased claim approval rates by 40%

Projects

Credit Card Fraud Detection Using GANs | *Python, TensorFlow, Scikit-learn, Pandas*

- Engineered a GAN based synthetic data augmentation pipeline for anomaly detection on a 284K transaction dataset with a 0.17% fraud rate, achieving F1 = 0.805 and 99.94% accuracy, outperforming SMOTE by 11.3% precision

InfoMate, an AI Powered University Information Chatbot | *Python, GPT-4o, Streamlit, MongoDB*

- Built an NLP powered chatbot on 10,000+ structured and unstructured documents using GPT-4o embeddings, validated with 10 domain expert faculty members, and deployed as a production-ready Streamlit app with MongoDB backed session management

Volunteer Experience

Event Volunteer **2025 to Present**
PMI Melbourne Chapter

- Coordinated check-in, setup and speakers and ran a session across 2 community events of 60+ attendees, and mentored new and existing members

Event Crew **2025**
RMIT Open Day

- Guided prospective students through campus and program information sessions

Professional Certifications

- Deloitte Australia, Data Analytics Job Simulation (Forage): Built data visualisations in Tableau and applied data analysis to business decision making scenarios
- Commonwealth Bank, Introduction to Data Science Job Simulation (Forage): Developed data analysis and visualisation solutions for financial decision making contexts

Technical Skills

Programming: Python, SQL, R, JavaScript
ML/AI: Prompt engineering, multi-model ensemble classification, LLM evaluation (GPT, Gemini, Claude), signal processing, SymPy validation engines
Data tools: Power BI, Tableau, Excel, MongoDB, Git and GitHub
Hardware/Systems: ESP32-S3 firmware flashing, IMU sensor validation, USB serial/Bluetooth LE streaming
Data analysis: Trend Analysis, Data Validation, Data Integrity, Reporting, EDA, Test automation (pytest), Statistical Analysis, Data Modeling, ETL
Soft skills: Cross-functional collaboration, stakeholder communication, technical documentation

Education

Master of Data Science **Feb 2025 – Dec 2026**
RMIT University, Melbourne *3.4/4.0*

- Coursework: Database Concepts, Big Data Management, Data Visualisation, Practical Data Science, Machine Learning, Time Series Analysis.

Bachelor of Technology **Aug 2016 – Nov 2020**
Jeppiaar Institute of Technology, India *3.12/4.0*

- Majors: Computer Systems, Information Systems